

Yeri Park

M.S. in Computer Science Student, Georgia Institute of Technology

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RESEARCH INTERESTS

My research lies at the intersection of computer architecture and systems, with a focus on hardware-software co-design for irregular sparse and graph computations. Specifically, I explore accelerator microarchitectures and runtime system optimizations on RISC-V GPGPU platforms. I am also deeply interested in the system-level challenges of efficient machine learning, investigating novel quantization, pruning, and scalable serving techniques to accelerate large-scale DNNs and LLMs/VLMs.

EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science

Atlanta, United States

Sep 2025 – May 2027 (Expected)

- *Related Courses:* GPU HW and SW, Advanced Operating Systems, HW-SW Co-Design, Deep Learning

Ewha Womans University

Bachelor of Science in Electronic and Electrical Engineering, GPA: 97.9/100 (Rank: 2nd in class)

Seoul, Republic of Korea

Mar 2017 – Feb 2022

RESEARCH EXPERIENCE

Georgia Institute of Technology

Graduate Research Intern, HPCA Lab (Advisor: Hyesoon Kim)

Atlanta, United States

Aug 2026 – Present

- Conduct graduate research on a RISC-V ISA-based open-source GPGPU architecture (Vortex), developing both the hardware and its compiler/runtime stack via SystemVerilog RTL, RTL simulation, and FPGA synthesis to evaluate hardware-software performance/area trade-offs
- Designing the Actor mechanism in hardware RTL to optimize irregular sparse computation, and building the accompanying runtime system software, to develop a dedicated hardware accelerator for graph computation

Hyundai Motor Company

Researcher, Department of Autonomous Driving SW Development

Seoul, Republic of Korea

Jan 2022 – Aug 2026

- Built and optimized the AI software running inside autonomous driving cars, making onboard computers process camera, radar, and LiDAR data faster while using less memory and power
- Reduced end-to-end network inference latency by 2.17x on production autonomous-driving DNNs by deploying mixed-precision quantization (PTQ/QAT), knowledge distillation, and structured pruning on NVIDIA Jetson (Thor/Orin) and Qualcomm Snapdragon; built the validation framework confirming recovered full-precision accuracy
- Accelerated VLM model inference by 1.6-1.7x using vLLM and TensorRT-LLM serving pipelines with automated layer-wise quantization and host ↔ GPU/NPU memory allocation across GPU server clusters and edge devices
- Delivered a production QNX/RTOS embedded AI inference framework (C/C++, CUDA, OpenCL) for autonomous systems, integrating car-network interfaces (LIN, CAN, Ethernet, LVDS), multimodal camera/LiDAR/Radar sensor logging, Rust-based visualization, and HILS/SILS validation across NVIDIA and Qualcomm platforms

Korea Institute of Science and Technology

Undergraduate Research Intern, Robot Vision Lab

Seoul, Republic of Korea

Sep 2021 – Dec 2021

- Researched and developed an RGB-Depth camera-based perception DNN model for a wearable assistive robot, running in real time on an embedded NVIDIA Jetson Xavier

Hyundai Motor Company

Undergraduate Research Intern, LiDAR Perception Development

Seoul, Republic of Korea

Jul 2021 – Sep 2021

- Built a real-time perception pipeline that lets a vehicle understand its surroundings from LiDAR sensor data, including detecting the ground, road edges, and nearby objects

Ewha Womans University

Undergraduate Research Intern, Digital Signal Processing Lab

Seoul, Republic of Korea

Jun 2019 – Sep 2019

- Developed a CNN detection model for contactless human breathing detection using IR-UWB radar signals

SKILLS

Programming: C/C++, Python (PyTorch), Rust, CUDA/OpenCL, SystemVerilog/RTL, Haskell, Git, GitHub, CI/CD

GPU & Compiler Tooling: TensorRT, TensorRT-LLM, vLLM, CUDA, ROCm, QNN, ONNX-MLIR, LLVM, RISC-V GPGPU (Vortex)

Model Optimization Techniques: PTQ/QAT, Knowledge Distillation, Structured/Unstructured Pruning, KV Cache Optimization

Embedded & RTOS: QNX, Linux Kernel, HILS/SILS Validation

Certifications: ISTQB SW Testing CTFL, ISTQB AI Testing, NVIDIA CUDA Programming

ADDITIONAL PROJECTS

SWE AI Agent GPU-CPU Load Time Optimization (Aug 2026) — Optimized GPU-CPU load time for an SWE (software engineering) agent workflow, profiling and restructuring data transfer and scheduling between host and device to reduce end-to-end agent task latency. *Georgia Institute of Technology*

Edge AI Network Pruning (Aug 2025) — Applied structured/unstructured pruning to CV networks on NVIDIA Orin. *C/C++, TensorRT, Swin Transformer*

Rust-Based ONNX-MLIR Optimization (Apr 2025) — Built a network optimization framework for universal hardware compilation. *Rust, Python, C++, ONNX-MLIR*

HONORS & AWARDS

2nd Prize, Ewha Engineering Capstone Design Contest (2020) — AI-based audience response analysis system

Excellence Award, Korea Electric Power Engineering Camp (2020) — Top 3, national engineering camp

Academic Excellence Scholarship, Ewha Womans University — 4 consecutive semesters (2017–2021)